Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **C++/Python systems**, large-data research pipelines, ML/NLP workflows, performance-tuned services, and production-grade platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-data research pipelines** across 100,000+ trajectory records, producing metrics, logs, reports, and reproducible artifacts for model behavior debugging, experiment comparison, statistical review, validation, and tuning.
- Integrated **HD-map geometry and trajectory context** with attention-based encoders, improving planning accuracy by **18%** while surfacing edge cases through simulation diagnostics, outlier analysis, reviews, reporting, and tuning.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **production data workflows** for 500+ users, translating requirements into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable delivery through reviewed changes and support.
- Improved **backend performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with diagnostic handoffs, reviews, and runbooks.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Python API and ETL services** using SQL, AWS Glue, and S3, improving response time by **30%** across integrations, release reviews, operational handoffs, troubleshooting, documentation, and supportability.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions during deployments, reviews, and checks.

Pandit Deendayal Energy University **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*

- Built a **statistical detection pipeline** over 650K+ URLs with 21 behavioral indicators, combining feature engineering, model evaluation, validation reports, outlier analysis, reproducible experiments, reviews, reporting, and audits.
- Achieved **96%+ detection accuracy** through systematic validation, feature comparison, and model diagnostics, documenting repeatable workflows for large-scale data review, statistical reporting, experiment handoffs, audits, reviews, and findings.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React dashboards and Python services** handling 10,000+ daily requests, implementing file ingestion, REST endpoints, automated checks, LLM workflows, documented fixes, production delivery, reviews, and testing.
- Prototyped **LLM-assisted evaluation workflows** with OCR, model routing, cost tracking, and validation cases, reducing feedback turnaround by **60%** through iterative debugging, user testing, diagnostics, reviews, and iteration.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash, HTML/CSS
Systems: data structures, algorithms, operating systems, Linux/Unix, distributed systems concepts, performance tuning
Data/ML: probability, statistics, machine learning, NLP, LLM workflows, PyTorch, Transformers, outlier analysis
Platforms: REST APIs, Flask, FastAPI, PostgreSQL, MySQL, MongoDB, AWS, GCP, Docker, CI/CD
Delivery: Git, Jenkins, PyTest, Mockito, code reviews, production debugging, documentation, resilient service design

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening algorithms, C++ runtime reasoning, debugging discipline, scheduler comparisons, profiling, and performance analysis.

Big-Brain Live | *FAISS, PostgreSQL, GCP, LLM APIs*

- Engineered an **LLM retrieval platform** indexing 50K+ documents with FAISS, PostgreSQL, and GCP deployment, reducing retrieval latency by **40%** through optimized ingestion, indexing, caching, and search workflows.

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*

- Benchmarked **transformer inference pipelines** across BLIP, GIT, and ViT-GPT-2 models on 10K images and 100K captions, improving caption accuracy by **15%** with diagnostics, validation, reporting, and testing.